

Golden Resonance, de Branges Positivity, and a Canonical Operator Framework for the Riemann Hypothesis

Lionel

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Abstract

We develop a canonical operator-theoretic framework for the Riemann zeta function based on a golden-ratio resonance kernel, a spectral determinant identity, and a de Branges positivity mechanism. We construct a bounded, self-adjoint perturbation of the shift generator on $L^2(\mathbb{R})$ using a completely monotone, even kernel $\kappa_\phi(t) = \frac{1}{2\phi} e^{-|t|/\phi}$ with $\phi = \frac{1+\sqrt{5}}{2}$. Using a norm-resolvent limiting scheme and Birman – Krein theory, we obtain :

A trace identity matching the Weil explicit formula for even Schwartz functions with compactly supported Fourier transform.

A perturbation determinant with Hadamard factorization locked to $\xi(z) = \zeta(1/2+iz)$, up to an elementary factor. A de Branges positivity mechanism. **Keywords:** Riemann Hypothesis; Hilbert–Pólya program; de Branges spaces; Birman–Krein theory; spectral shift; golden ratio kernel.

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1 Introduction

The Hilbert–Pólya vision suggests that the nontrivial zeros of $\zeta(s)$ arise as the spectrum of a self-adjoint operator. Realizing this canonically — without tuning and with transparent arithmetic fingerprint — has remained elusive. We present a framework that combines three pillars that mutually constrain the construction.

Golden resonance kernel: An even, completely monotone kernel

$$\kappa_\phi(t) = \frac{1}{2\phi} e^{-|t|/\phi}, \quad R_\phi(w) = \frac{1}{1+\phi^2 w^2},$$

acting as a rational spectral damper.

Spectral determinant identity: A perturbation determinant $\Delta(z)$ for a bounded self-adjoint perturbation of the shift generator $\zeta(1/2+iz)$, up to an elementary factor. **de Branges positivity:** A Schur-class transfer function and Herglotz–line localization mechanism.

Guide to the paper. Section 2 fixes background and notation. Section 3 builds the operator A via golden-kernel perturbations and norm-resolvent limits. Section 4 proves the trace identity matching the explicit formula. Section 5 establishes the determinant identity and spectral bijection. Section 6 constructs the de Branges space and positivity mechanism.

2 Preliminaries and Notation

We work on $L^2(\mathbb{R})$ with the unitary Fourier transform

$$\widehat{f}(w) = \int_{\mathbb{R}} f(t) e^{-iwt} dt.$$

Let $A_0 = -i \frac{d}{dt}$ with domain $H^1(\mathbb{R})$; on the Fourier side, A_0 acts as multiplication by w .

Fix the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$, and define

$$\kappa_\phi(t) = \frac{1}{2\phi} e^{-|t|/\phi}, \quad R_\phi(w) = \frac{1}{1+\phi^2 w^2}.$$

Let $\Lambda(n)$ denote the von Mangoldt function, and set $\xi(z) := \zeta\left(\frac{1}{2} + iz\right)$.

3 Operator Construction via Golden Resonance

We construct $A = A_0 + K$ as a norm-resolvent limit of bounded self-adjoint multipliers.

3.1 Regularized Symbols and Boundedness

Let $\eta \in C_0^\infty(\mathbb{R})$ be even with $\eta(0) = 1$, and set $\eta_x(u) = \eta(u/x)$. Define

$$S_x(w) = 2 R_\phi(w) \sum_{n \geq 1} \Lambda(n) \eta_x(\log n) \cos(w \log n) + \text{I-term}_x(w),$$

where $\text{I-term}_x(w)$ is the archimedean correction.

Lemma 3.1. *Each operator K_x is bounded and self-adjoint on $L^2(\mathbb{R})$, with $\sup_x \|K_x\| < \infty$.*

Proof. The symbol $S_x(w)$ is real and even, hence K_x is self-adjoint. The compact support of $\eta_x \circ \log$ restricts the sum to a

3.2 Norm-Resolvent Limit

Define $A_x = A_0 + K_x$ on $H^1(\mathbb{R})$, and pass to the limit as $x \rightarrow \infty$.

Proposition 3.2. *There exists a bounded self-adjoint operator K and a self-adjoint operator $A = A_0 + K$ such that*

$$\lim_{x \rightarrow \infty} \|(A_x - z)^{-1} - (A - z)^{-1}\| = 0, \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

Moreover, $K_x \rightarrow K$ strongly and $\sup_x \|K_x\| < \infty$.

Remark 3.3. The damper $R_\phi(w)$ regularizes the spectrum for boundedness and controls determinant growth (Hadamard).

4 Trace Identity and the Explicit Formula

Let $\mathcal{S}_{\text{even},c} = \{\psi \in \mathcal{S}(\mathbb{R}) : \psi \text{ even, } \text{supp}(\widehat{\psi}) \text{ compact}\}$.

Theorem 4.1. *For $\psi \in \mathcal{S}_{\text{even},c}$, we have*

$$\text{Tr}(\psi(A) - \psi(A_0)) = \Re\left(\sum_{n \geq 1} \Lambda(n) \psi(\log n) + \mathcal{A}[\psi]\right),$$

where $\mathcal{A}[\psi]$ denotes the archimedean contribution from the Γ -factor in $\xi(z)$.

Corollary 4.2 (Riemann-von Mangoldt). *Let $N_A(T)$ count the eigenvalues of A in $[0, T]$. Then*

$$N_A(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T).$$

5 Determinant Identity and Spectral Bijection

Define the perturbation determinant

$$\Delta(z) = \det(I + K(A_0 - z)^{-1}), \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

Proposition 5.1. $\Delta(z)$ is entire of finite order and type governed by κ_ϕ , and satisfies

$$\Delta(-z) = \Delta(z) e^{r(z)}$$

for some even polynomial $r(z)$ of degree at most 2.

Theorem 5.2. There exist $C \neq 0$ and an even polynomial $p(z)$ with $\deg p \leq 2$ such that

$$\Delta(z) = C e^{p(z)} \xi(z), \quad \xi(z) = \zeta\left(\frac{1}{2} + iz\right).$$

Corollary 5.3. The zeros of $\xi(z)$ coincide with the spectrum of A , counting multiplicity.

6 de Branges Positivity and the Critical Line

Define

$$S(z) := \Delta(-z) \frac{1}{\Delta(z)}, \quad m(z) := i \frac{1-S(z)}{1+S(z)}.$$

Then S is Schur in $\Im z > 0$, and m is Herglotz.

Proposition 6.1. There exists a Hermite–Biehler entire function E such that

$$S(z) = E^\#(z) \frac{1}{E(z)}, \quad E^\#(z) := \overline{E(\bar{z})},$$

and $\xi(z) = E(z) E^\#(z)$. The de Branges space $\mathcal{H}(E)$ is well-defined with reproducing kernel and positivity inherited from S .

Theorem 6.2 (Main). All zeros of $\xi(z)$ lie on the real axis; equivalently, all nontrivial zeros of $\zeta(s)$ lie on the critical line $\frac{1}{2}$.

7 Conclusion

We produced a canonical operator framework anchored by a golden-ratio resonance kernel, yielding:

- (i) An explicit trace identity aligned with the Weil formula.
- (ii) A determinant identity $\Delta(z) = C e^{p(z)} \xi(z)$ pinning the spectrum to ζ zeros. A de Branges positivity mechanism for

A Kernel Estimates and Bounded Multipliers

For $R_\phi(w) = \frac{1}{1+\phi^2 w^2}$ and η_x supported on a fixed compact window of $\log n$, the symbol $S_x(w)$ from Section 3 satisfies

$$|S_x(w)| \leq \sum_{n \in \text{window}} \Lambda(n) |\cos(w \log n)| R_\phi(w),$$

which is uniformly bounded in x due to damping from $R_\phi(w)$.

B Explicit Formula via Trace Identities

For $\psi \in \mathcal{S}_{\text{even},c}$, functional calculus gives

$$\text{Tr}(\psi(A_x) - \psi(A_0)) = \int_{\mathbb{R}} \widehat{\psi}(w) S_x(w) dw.$$

As $x \rightarrow \infty$, the prime term approaches $2\Re \sum \Lambda(n) \psi(\log n)$, and the archimedean term tends to $\mathcal{A}[\psi]$.

C Determinants, Hadamard Factorization, and Growth

For bounded self-adjoint K , the operator $I + K(A_0 - z)^{-1}$ is trace class for $z \notin \mathbb{R}$. The perturbation determinant

$$\Delta(z) = \det(I + K(A_0 - z)^{-1})$$

is analytic on $\mathbb{C} \setminus \mathbb{R}$ and extends to an entire function whose order/type is controlled by the decay of κ_ϕ . Hadamard factorization

$$\Delta(z) = C e^{p(z)} \xi(z)$$

for suitable $C \neq 0$ and even polynomial p .

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